AP PRECALCULUS

Unit 1 • Lesson 1.11 practice

Lesson 1.11 • Equivalent Representations of Polynomial and Rational Expressions

5 multiple-choice questions and 1 free-response question

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

Which is the standard form of $(x+1)^3$?

(A) x^3+1
(B) x^3+3x^2+3x+1
(C) x^3+3x+1
(D) x^3+2x^2+2x+1

Question 2

NO CALCULATOR

What is the remainder when x^3-2x+5 is divided by $x-2$?

(A) 1
(B) 5
(C) 9
(D) 13

Question 3

NO CALCULATOR

Which is the completely factored form of x^3-9x?

(A) $x(x-3)(x+3)$
(B) $x(x-3)^2$
(C) $(x-3)(x+3)$
(D) $x(x^2+9)$

Question 4

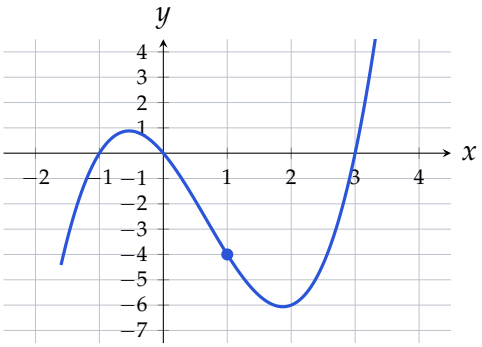
NO CALCULATOR

What is the coefficient of x^3 in the expansion of $(x+2)^5$?

(A) 10
(B) 20
(C) 80
(D) 40

Question 5 (graph) NO CALCULATOR

The graph of a cubic polynomial p with leading coefficient 1 is shown. Its zeros are -1, 0 and 3. Which is $p(x)$ in standard form?



(A) $x^3 - 2x^2 - 3x$
(B) $x^3 + 2x^2 - 3x$
(C) $-x^3 + 2x^2 + 3x$
(D) $x^3 - 3x$

Question 6 (Free response) NO CALCULATOR

Let $p(x) = 2x^3 + x^2 - 13x + 6$.

(a) Show that $x = 2$ is a zero of p.
(b) Write $p(x)$ as a product of linear factors and find all zeros of p.
(c) Write $\dfrac{p(x)}{x+1}$ in the form $q(x) + \dfrac{k}{x+1}$.